

SUMMARY

AI research scientist and engineer with a Ph.D. in multimodal deep learning, 10+ publications, and 5 U.S. patents, combining peer-reviewed research with 4+ years of experience taking LLM systems from foundational research to production.

Research interests: LLM post-training (SFT, GRPO/RLHF), agentic AI safety and red-teaming, multimodal learning.

EDUCATION

University of Arizona

Ph.D., **Focus in Multimodal Deep Learning**

2024

M.S. in Computer Science, **Focus in Natural Language Processing**

2022

M.S. in Systems and Industrial Engineering, **Focus in Machine Learning**

2018

TECHNICAL STRENGTHS

LLM Post-Training

SFT, GRPO/RLHF, PEFT (LoRA), Red-Teaming, Evals & Benchmarking

AI & ML Expertise

Multimodal Learning, Reinforcement Learning, NLP, Agentic AI, Time Series

Deep Learning Frameworks

PyTorch, HuggingFace (Transformers, TRL, PEFT), PyTorch Lightning, TensorFlow

Computer Programming

Python, Java, Scala, SQL, Bash

Software & Tools

Docker, Kubernetes, Git, Databricks, AzureML

WORK EXPERIENCE

Lead AI Engineer, Armada AI

Aug 2023 - Present

- **Pioneered AI safety and red-teaming research** on open-weight LLM supply-chain attacks: designed a multi-stage PEFT attack (SFT-then-GRPO) that implants conditional backdoors in tool-using agents (**99.6%** attack success while evading standard benchmarks) and a layered defense framework achieving **100% runtime detection** (under review, NeurIPS 2026).
- **Architected** a scalable agentic AI routing system optimized for edge deployment, utilizing advanced techniques like GRPO to outperform large-scale proprietary models by **over 10%**.
- **Spearheaded** the end-to-end lifecycle—from foundational research to production deployment—of a prescriptive intelligence platform. Directed cross-functional engineering and product teams to deliver actionable, LLM-driven insights, directly securing a **\$1M+ enterprise contract** and improving projected operational efficiency by **4.5%**.
- **Developed** a natural language interface for robotic systems, improving task accuracy by **over 20%** and leading to a granted U.S. patent and publications at HRI 2026 and ICCAR 2025.
- **Led** a large-scale geospatial analysis of LEO satellite network performance, predicting service degradation from severe weather (published at ACM SIGSPATIAL 2025).
- **Manage a team of 4 AI engineers** while serving as technical lead across multiple high-impact projects; mentor engineers and interns, and recognized with the company's **Anchor Award** for going above and beyond in execution and customer-centric problem solving.

Data Scientist, American Airlines

Aug 2022 - Aug 2023

- **Developed a deep neural network forecasting engine** that predicts flight-level traffic for the Yield Management team, reducing prediction mean squared error by 50% through the use of multimodal neural network architecture and pre-training on historical data, including time series and multi-dimensional data (arXiv preprint).
- **Led** the development of the forecasting engine and spearheaded its deployment into the airline system for **A/B testing** and potential replacement of older methods. Worked closely with the IT team on the project development using AzureML and **trained and supervised** a team of four in the use and evaluation of the approach.
- **Expanded** the analysis to the entire market of ~7,000 flights, forecasting 330 days into the future. Demonstrated superior performance in capturing trends, seasonality, and sudden capacity-driven demand shifts that previous models missed.

Machine Learning Postdoctoral Researcher, Truveta

May 2022 - Aug 2022

- **Pre-trained and fine-tuned Large Language Models (LLMs)** on clinical text data, resulting in improved overall accuracy for upstream and downstream tasks and significantly increased downstream training and inference speeds (fourfold and twofold increase, respectively).
- **Achieved state-of-the-art results** in the ontology alignment task, with at least a **5% improvement** in F1, Hit@1, and MRR. Developed a novel framework (Truveta Mapper) using a multi-task sequence-to-sequence LLM for unsupervised ontology alignment, outperforming existing solutions in runtime and alignment quality (published at OM @ ISWC 2023; U.S. patent pending).

- **Implemented** a novel plot-processing approach utilizing object detection models and text extraction techniques, correctly identifying data from plots at a rate **approaching the state-of-the-art** (~69%). Deployed on the RedShred API for automatic processing of PDF documents.
- **Utilized** HuggingFace framework to develop and train transformer-based methods for document understanding, including segmentation and object detection. These techniques demonstrated strong performance in the field of document analysis.

- **Developed and designed a novel multimodal deep learning architecture** that employs reinforcement learning to dynamically adjust network depth, achieving an approximate **10-15% reduction in mean squared error (MSE)** across diverse tasks compared to SOTA models (BiDepth; DeepShallow, CIE51).
- **Introduced a novel Textual Question Answering architecture** that employs on-demand visual grounding to enhance answer accuracy. The 400M parameter model demonstrates **comparable performance to significantly larger models like GPT-4o** (published in ACM TOMM).

PUBLICATIONS

- Ehsani, S.**, Pallakonda, B., Hindsbo, M., and Mishra, P., 2026. The Poisoned Toolbox: Backdoor Attacks and Defenses for Tool-Using Agents in Open-Weight LLMs. *Under review at NeurIPS 2026*.
- Pallakonda, B., Hindsbo, M., **Ehsani, S.** and Mishra, P., 2026. Sleeper cell: Injecting latent malice temporal backdoors into tool-using llms. *arXiv preprint arXiv:2603.03371*.
- Hindsbo, N., **Ehsani, S.**, and Mishra, P., 2026. SCOPE: Real-Time Natural-Language Camera Control on the Edge. *In ACM/IEEE HRI 2026*.
- Ehsani, S.**, Pallakonda, B., and Mishra, P., 2025. Mapping the Storm: Geospatial Impacts of Severe Weather on LEO Network Performance. *In ACM SIGSPATIAL 2025*.
- Ehsani, S.**, Pan, F., Hu, Q. and Liu, J., 2025. BiDepth Multimodal Neural Network: Bidirectional Depth Deep Learning Architecture for Spatial-Temporal Prediction. *arXiv preprint arXiv:2501.08411*.
- Buynitsky, A., **Ehsani, S.**, Pallakonda, B., and Mishra, P., 2025. Camera Control at the Edge with Language Models For Scene Understanding. *In IEEE ICCAR 2025 Proceedings*.
- Ehsani, S.**, and Liu, J., 2024. From the Depths to the Surface: Navigating Spatial Temporal Data with DeepShallow Network. *In CIE51 Proceedings*.
- Ehsani, S.** and Liu, J., 2024. Elevating Textual Question Answering with On-Demand Visual Augmentation. *ACM Transactions on Multimedia Computing, Communications and Applications*.
- Ehsani, S.**, Sergeeva, E., Murdy, W. and Fox, B., 2024. Predicting the Skies: A Novel Model for Flight-Level Passenger Traffic Forecasting. *arXiv preprint arXiv:2401.03397*.
- Amir, M., Baruah, M., Eslamialishah, M., **Ehsani, S.**, Bahramali, A., Naddaf-Sh, S. and Zarandioon, S., 2023. Truveta Mapper: A Zero-shot Ontology Alignment Framework. *In OM-2023: 18th International Workshop on Ontology Matching, ISWC 2023*.
- Ehsani, S.**, Reddy, C.K., Foreman, B., Ratcliff, J. and Subbian, V., 2021. Subspace Clustering of Physiological Data From Acute Traumatic Brain Injury Patients: Retrospective Analysis Based on the PROTECT III Trial. *JMIR Biomedical Engineering*, 6(1), p.e24698.

PATENTS

- Mishra, P.K., Buynitsky, A., and **Ehsani, S.**, Armada Systems Inc., 2025. *Robotic control using natural language commands*. U.S. Patent 12,289,517.
- Ehsani, S.**, and Mishra, P.K., Armada Systems Inc., 2024. *Natural language statistical model with alerts*. U.S. Patent 12,086,557.
- Ehsani, S.**, and Mishra, P.K., Armada Systems Inc., 2024. *Time series data to statistical natural language interaction*. U.S. Patent 12,067,041.
- Ehsani, S.**, and Mishra, P.K., Armada Systems Inc., 2025. *Natural language statistical model with workspaces*. U.S. Patent Application (pending).
- Marcjan, C.A., Amir, M., Baruah, M., Eslamialishah, M., **Ehsani, S.**, Bahramali, A., Naddaf-Sh, S., and Zarandioon, S., Truveta Inc., 2023. *Systems and methods for ontology matching*. U.S. Patent Application 20240087687 (pending).